

AI-POWERED STARTUP

AutoIntern

Complete Project Blueprint

Architecture | Build Plan | GTM Strategy | Monetization

1. Executive Summary

AutoIntern is an AI-powered internship application engine that finds relevant openings, tailors each resume and cover letter to the specific job description, and helps students apply to 20+ positions in under 10 minutes. The product starts on college campuses as a proven distribution channel and scales nationally to every job-seeking student in India.

The problem: Students spend 3-4 hours per application manually customizing resumes and cover letters. Response rates from generic applications hover below 2%. Meanwhile, thousands of internship seats go unfilled because companies receive irrelevant, copy-paste applications.

The solution: AutoIntern's AI reads a JD in seconds, rewrites the student's resume bullets to match keywords and requirements, generates a personalized cover letter, finds the right HR contact email, and queues up applications for one-click sending. Students go from 2 applications a week to 20 a day.

Metric	Target
Target market (India)	8 crore+ college students
Addressable students seeking internships	2 crore+ students
Campus launch (Month 1)	1 college, 5,000 students
Monetization start	Month 2 (freemium to paid)
Revenue target (Year 1)	Rs. 50 lakh ARR
Revenue target (Year 2)	Rs. 3 crore ARR

2. Product Overview

2.1 Core User Flow

The product is designed for zero friction. A student should be able to go from signup to their first batch of applications in under 15 minutes.

- **Step 1 - Onboarding:** Student signs up with college email for trust verification. Uploads resume (PDF/Word). Fills a 2-minute profile (skills, preferred roles, location, stipend expectation).
- **Step 2 - Job Discovery:** System aggregates internship listings from public sources (company career pages, RSS feeds, allowed APIs). AI ranks them by match score against student profile.
- **Step 3 - AI Tailoring:** For each job, AI rewrites resume bullet points using JD keywords, generates a customized cover letter (150-200 words), and scores the match probability.
- **Step 4 - Application Queue:** Student sees a dashboard of tailored applications ready to send. Reviews, edits if needed, clicks Send. Bulk send up to 20 at once.
- **Step 5 - Tracking:** Dashboard tracks sent applications, opens (email tracking pixel), replies, and interview status. AI suggests follow-up messages after 5 days of no response.

2.2 Key Features

Feature	Description
Resume Tailor	AI rewrites bullet points matching JD keywords. Preserves truth, enhances relevance.
Cover Letter Gen	Personalized 150-200 word letters using company name, role, and student's specific projects.
Job Aggregator	Pulls from 50+ public sources. Filters by role, stipend, location, work-mode.
Match Score	0-100 score showing how well student's profile fits each JD before applying.
Email Outreach	Finds HR email via Hunter.io API. Drafts cold outreach for companies with no portal.
Application Tracker	Kanban board: Applied > Seen > Replied > Interview > Offer.
Follow-up AI	Auto-drafts polite follow-up emails after configurable delay (5-7 days).
Analytics	Weekly report: applications sent, response rate, best-performing resume versions.

3. Technical Architecture

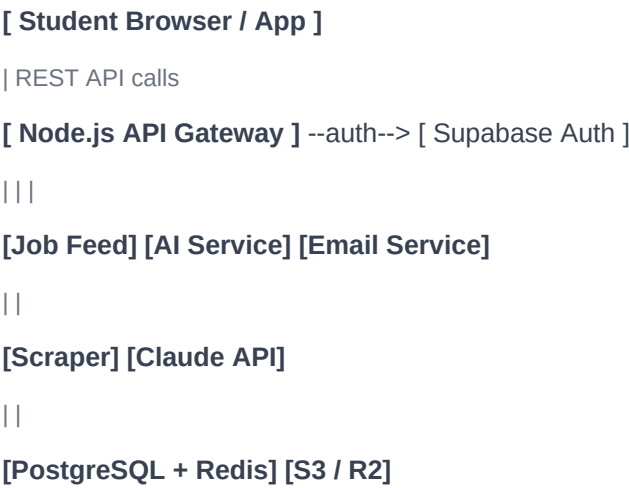
3.1 High-Level Stack

AutoIntern is a full-stack web application with a React frontend, Node.js + Python backend, and AI inference via Anthropic/OpenAI APIs. The architecture is modular - each major feature (scraper, AI tailoring, email, tracker) is an independent service.

Layer	Technology	Purpose
Frontend	React + TypeScript	Student dashboard, job feed, application tracker, resume editor
Backend API	Node.js (Express)	Auth, user management, job feed API, application queue, webhooks
AI Service	Python (FastAPI)	Resume tailoring, cover letter gen, match scoring, follow-up drafts
Scraper Service	Python (Scrapy/Playwright)	Job aggregation from career pages, RSS, public APIs
Database	PostgreSQL	Users, resumes, jobs, applications, analytics
Cache	Redis	Job feed cache, rate limiting, session store
File Storage	AWS S3 / Cloudflare R2	Resume PDFs, generated documents
Email	Resend / AWS SES	Application emails, follow-ups, notifications
Auth	Supabase Auth	College email verification, OAuth (Google)
Hosting	Railway / Render	Backend + AI service (easy deploy, cheap start)
CDN + Frontend	Vercel	React app, edge caching
AI APIs	Anthropic Claude API	Resume rewriting, cover letter, match scoring

3.2 System Architecture Diagram

Below is the data flow across all major components:



Simplified architecture - each box is an independent deployable service

3.3 Database Schema (Key Tables)

Table	Key Columns
users	id, email, college_id, name, created_at, plan
resumes	id, user_id, raw_text, parsed_json, file_url, version
jobs	id, title, company, jd_text, source_url, stipend, location, scraped_at
applications	id, user_id, job_id, tailored_resume_url, cover_letter, status, sent_at
email_outreach	id, application_id, to_email, subject, body, opened_at, replied_at
analytics	id, user_id, week, apps_sent, response_rate, interviews

4. Build Plan — Phase by Phase

Phase 1: MVP (Weeks 1-4)

Goal: Get one student to send 10 tailored applications using the product. Nothing else matters.

- **Week 1** - Setup: Repo, auth (Supabase), basic React dashboard shell, PostgreSQL schema, deploy skeleton on Railway.
- **Week 2** - Resume Parser + AI Tailor: Accept PDF resume upload. Parse text. Call Claude API with resume + JD. Get back tailored bullets + cover letter. Display to user.
- **Week 3** - Job Scraper v1: Scrape 5 target sites (Internshala public listings, LinkedIn public, company career pages). Store in DB. Show feed to user.
- **Week 4** - Application Queue + Send: User selects jobs, sees tailored app, clicks Send. Email sent via Resend API. Basic tracker (Applied/Replied columns).

Phase 2: Polish + Retention (Weeks 5-8)

- **Week 5** - Match Scoring: Score each job 0-100 against student profile. Sort feed by score. Students only see relevant jobs.
- **Week 6** - Tracker + Follow-ups: Kanban tracker. Auto-schedule follow-up email draft after 5 days. Email open tracking pixel.
- **Week 7** - Analytics Dashboard: Weekly stats (apps sent, response rate, best resume version). A/B test two resume versions.
- **Week 8** - Onboarding polish + referral: Smooth first-time flow. Referral link (invite a friend, get 10 extra free apps/month).

Phase 3: Scale (Month 3+)

- Scraper v2: Add 50+ sources, auto-detect new company career pages.
- Mobile app: React Native wrapper for notifications (interview calls come fast).
- College admin dashboard: Sell placement data to TPO (Training & Placement Officers).
- B2B: Companies pay to get pre-filtered, AI-ranked applicant lists from AutoIntern.

Phase	Timeline	Milestone
MVP	Week 1-4	First student sends 10 applications via AutoIntern
Polish	Week 5-8	100 active users, 5% weekly retention
Campus Launch	Month 3	1,000 users on home campus, first paying users
Monetization	Month 4	500 paid subscribers @ Rs. 199/month = Rs. 1L MRR

Expansion	Month 6	5 campuses, 10,000 users, Rs. 5L MRR
Series A prep	Month 12	50,000 users, Rs. 50L ARR, B2B pipeline started

5. AI Implementation — How to Build It

5.1 Resume Parser

Extract structured data from raw resume PDF text using Claude:

```
prompt = f'''Extract from this resume:\n{resume_text}\nReturn JSON with: name, email, skills[], experience[], education[], projects[]. Return only valid JSON.'''
```

5.2 Resume Tailor Prompt

The core AI call. Feed resume + JD, get back rewritten bullets:

```
prompt = f'''You are an expert resume writer.\nJob Description:\n{jd_text}\n\nStudent Resume:\n{resume_json}\n\nRewrite the experience and project bullet points to match the JD keywords. Keep facts true. Make bullets start with strong action verbs. Return JSON with updated bullets only.'''
```

5.3 Cover Letter Generation

```
prompt = f'''Write a 180-word cover letter for {student_name} applying to {role} at {company}.\nTheir strongest relevant experience: {top_2_bullets}\nJD summary: {jd_summary}\nTone: confident, specific, not generic. No cliches.'''
```

5.4 Match Scoring

```
prompt = f'''Score this student profile vs job description from 0-100.\nProfile skills: {skills}\nJD required skills: {required_skills}\nReturn only a JSON: {{score: int, reason: string (1 sentence)}}'''
```

5.5 Scraping Strategy (Legal)

Source Type	Approach
Company career pages	Scrapy spider crawls /careers, /jobs paths. Respects robots.txt. Caches 24 hrs.
Internshala public	Public listing pages (no login needed). Parse HTML. Rate limit: 1 req/3 sec.
RSS feeds	Many job boards publish RSS. Parse with feedparser. Zero risk, fully allowed.
LinkedIn public	Public job posts only (no login). Use Playwright for JS-rendered pages.
Hunter.io API	Find HR contact emails for cold outreach. 50 free lookups/month on free tier.

6. Monetization Strategy

6.1 Pricing Tiers

Plan	Details
Free	10 AI-tailored applications/month. Basic tracker. No follow-ups.
Pro - Rs. 199/month	Unlimited applications. Follow-up automation. Analytics. Match scoring.
Pro+ - Rs. 499/month	Everything in Pro + cold email outreach + resume A/B testing + priority support.
College License - Rs. 25,000/year	Entire campus access. Placement analytics for TPO. Co-branded.

6.2 Revenue Projections

Month	Users	Revenue
Month 2	200 free, 50 paid	Rs. 9,950/month
Month 4	1,000 free, 300 paid	Rs. 59,700/month
Month 6	5,000 free, 1,000 paid	Rs. 2,50,000/month
Month 12	20,000 free, 4,000 paid	Rs. 10,25,000/month

6.3 B2B Revenue (Month 6+)

- Companies pay Rs. 5,000-15,000/month to post directly to AutoIntern and receive AI-ranked applicant shortlists.
- Placement agencies pay for bulk access to student profiles (anonymized until student opts in).
- Coaching institutes white-label AutoIntern for their students as a premium feature.

7. Go-to-Market Strategy

7.1 Campus Launch Playbook

- **Week 0:** Build waitlist landing page. Collect 200 signups via WhatsApp, Instagram, college Facebook groups before launch.
- **Launch Day:** Stall in canteen/main block. Demo live: show student their resume getting tailored in 30 seconds. Free Pro for first 100 signups.
- **Week 1-2:** Recruit 5 campus ambassadors (one per department). They get 3 months Pro free + Rs. 50 per paid referral.
- **Week 3+:** WhatsApp group sharing. 'I got 3 interview calls in a week' testimonials. Placement season timing is critical.

7.2 Timing is Everything

Placement season (Aug-Oct, Jan-Mar) is when students panic. Launch 3 weeks before placement season starts. That's your Black Friday.

7.3 Expansion Playbook

- After campus 1 works, identify 4 more colleges in the same city. Use ambassador from campus 1 who knows seniors in other colleges.
- Partner with college placement cells (TPO). Give them a free dashboard showing placement analytics. They promote you in return.
- Target engineering colleges first (highest density of internship seekers + highest willingness to pay for career tools).

7.4 Growth Channels

Channel	Strategy
WhatsApp	Share 'I applied to 20 companies in 20 minutes' screenshots. Viral by nature.
Instagram Reels	30-sec demo video: resume before vs after AI tailoring. Show the transformation.
College Fests	Sponsor hackathon prize. Host 'resume roast' session (bring resume, AI fixes it live).
LinkedIn	Students post about AutoIntern after getting interviews. Organic B2B discovery.
SEO	'Best internship sites India 2026' content. Long-term free traffic.

8. Team Structure & Execution

8.1 Minimum Team for MVP

Role	Responsibilities
Full Stack Lead (you)	Architecture decisions, backend API, database, deployment, AI integration
Frontend Dev	React dashboard, job feed UI, application queue, tracker
Scraper/Data Eng	Job scrapers, data pipeline, source expansion
Growth/Marketing	Campus ambassador program, social media, partnerships

8.2 Tools & Cost to Launch

Tool	Cost
Vercel (frontend hosting)	Free tier
Railway (backend hosting)	~Rs. 800/month
Supabase (auth + DB)	Free tier (up to 50k users)
Anthropic Claude API	~Rs. 4,000/month (1000 tailoring calls)
Resend (email sending)	Free tier (3,000 emails/month)
Hunter.io (email finder)	Free tier (50 lookups/month)
Cloudflare R2 (file storage)	Free tier (10GB)
Total burn (Month 1-2)	~Rs. 5,000-8,000/month

8.3 Risks & Mitigations

Risk	Mitigation
Job portals block scraper	Focus on company career pages + RSS (unblockable). 50+ sources = redundancy.
Low conversion free to paid	Paywall hits at 10 apps. Placement season = high willingness to pay.
AI tailoring quality is bad	Human review mode: student edits before sending. AI improves with feedback.
Competitor copies idea	Moat is data (student profiles + job match history) + campus trust network.
College bans the app	Never automate portal logins. Stay legal. Build relationships with TPO.

Final note: The product that wins is not the one with the most features. It is the one that gets a student their first interview call. Build toward that single outcome first. Everything else follows.

